

CPDet3D

Learning Class Prototypes for Unified Sparse Supervised 3D Object Detection
The IEEE/CVF Conference on Computer Vision and Pattern Recognition(CVPR) 2025

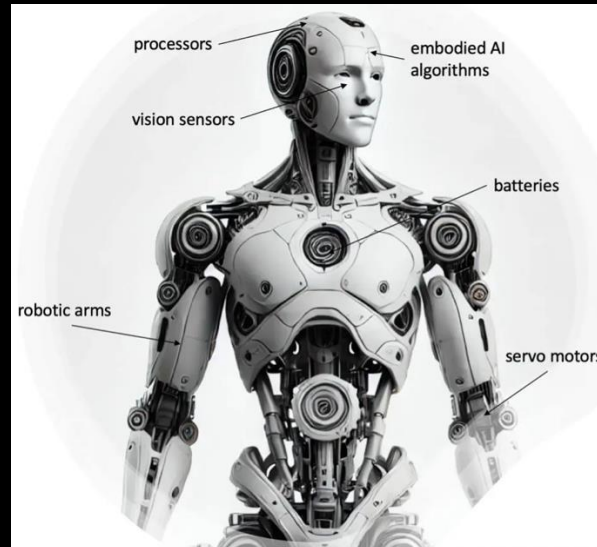
허정원 이찬영 이진주 황장훈 안가영

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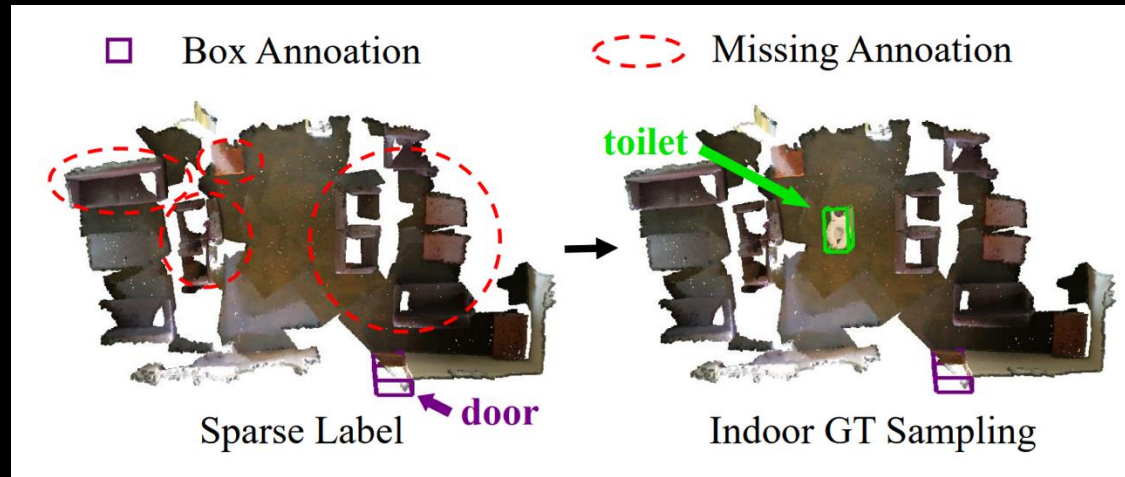
Introduction

- Both **indoor** and **outdoor** scene perceptions are essential for **embodied intelligence**.



- However, current sparse supervised 3D object detection methods focus solely on **outdoor scenes**.

Introduction – indoor scene



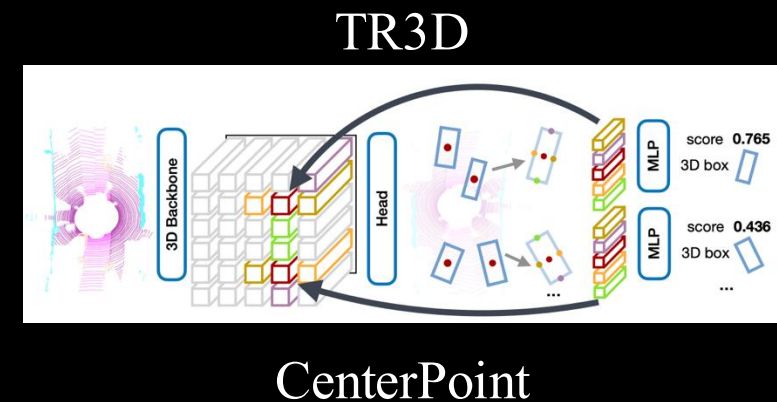
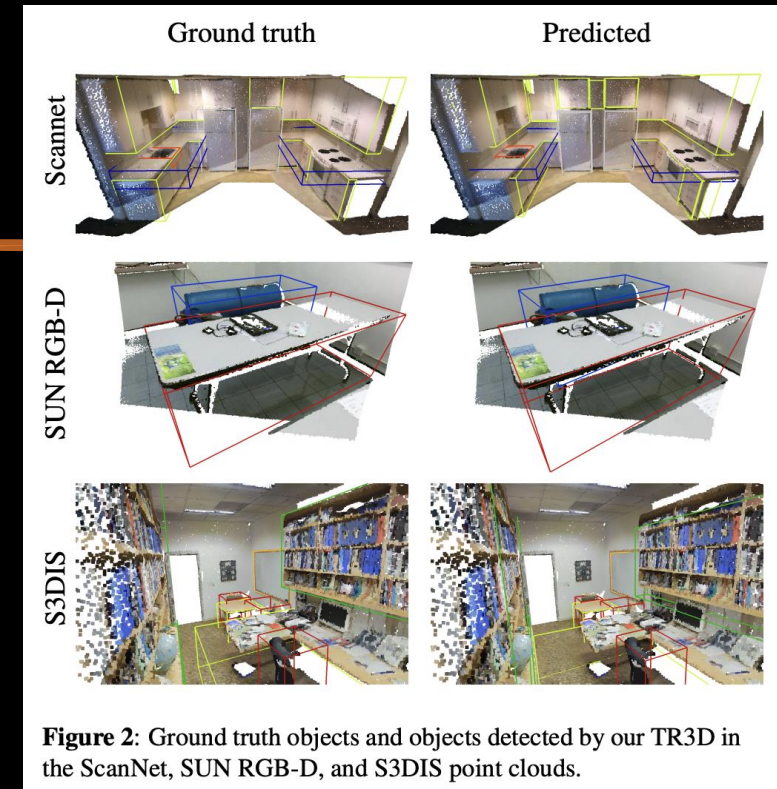
Visualization of GT sampling on indoor dataset ScanNet V2.

- Sparse supervision reduces annotation cost by labeling only one object per scene.
- However, existing methods rely on GT sampling to ensure full category coverage per scene — a condition often violated in indoor settings.

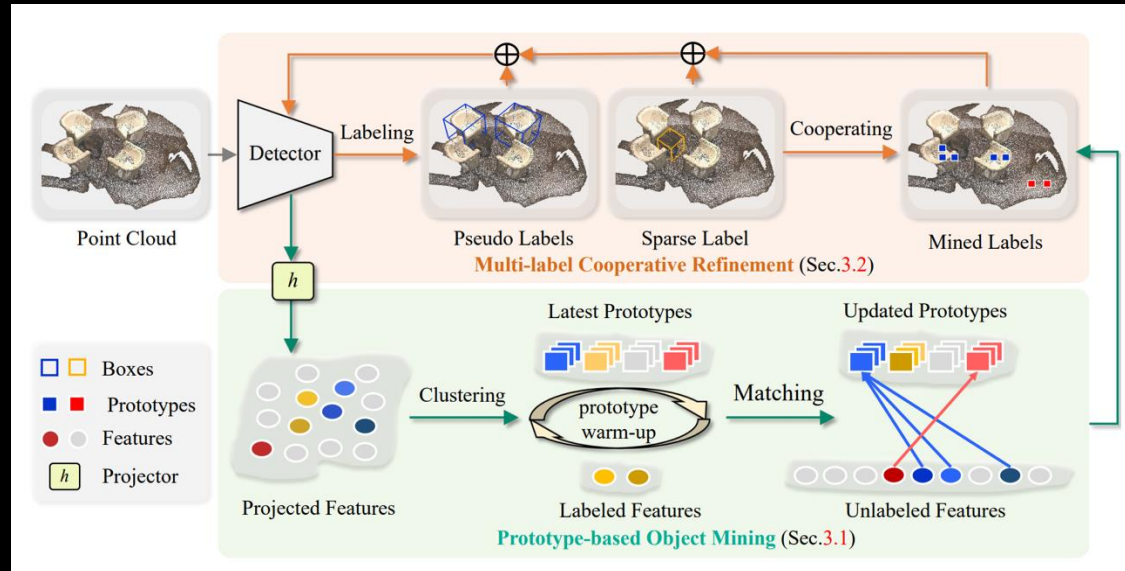
→ This work propose to mine unlabeled objects via **class prototype matching** and **refine** detection using cooperative pseudo and prototype labels.

Related Work

- **TR3D** improves the results of FCAF3D with a simpler structure and is widely used in indoor 3D object detection.
- Used as the base detector for indoor datasets due to its competitive **accuracy** and **simplicity**.
- **CenterPoint** is an efficient anchor-free one-stage detector that regresses object boxes based on predicted center points.
- Used as the base detector for outdoor datasets due to **compatibility** and **scalability**.

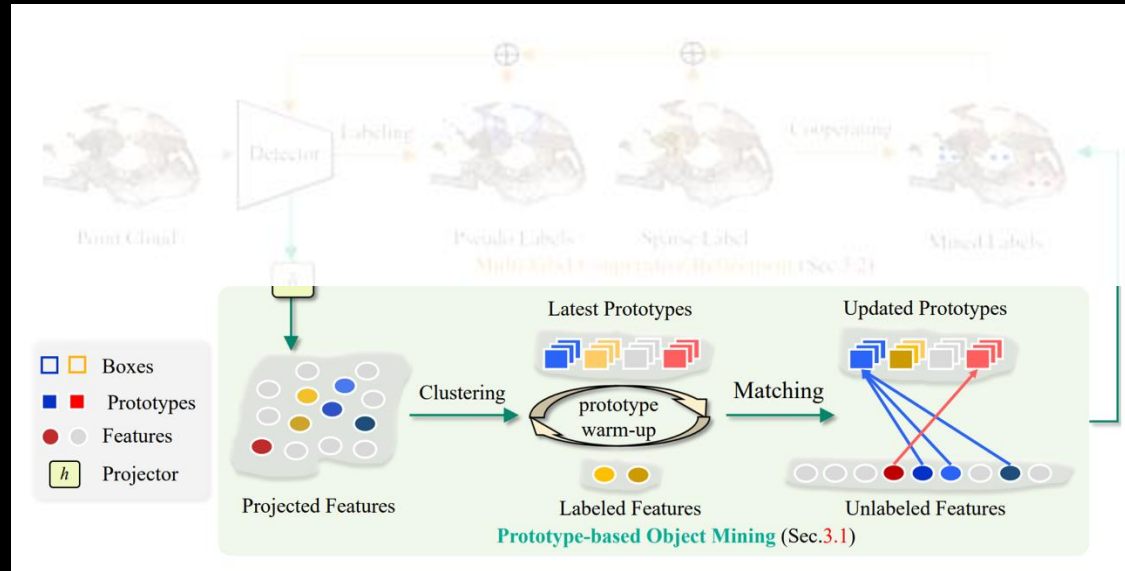


Methods



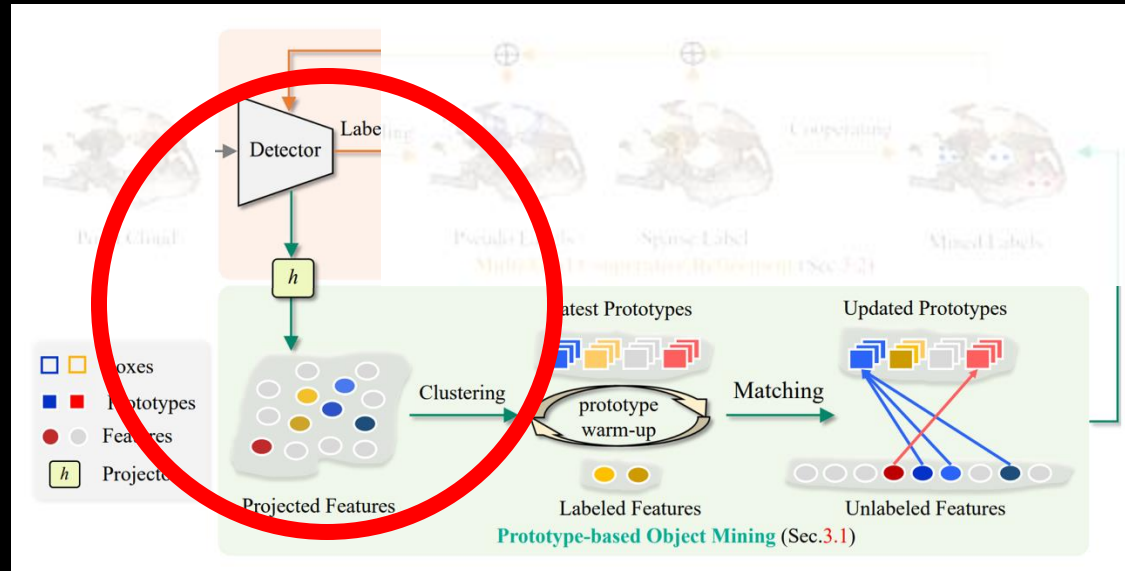
- **First**, this work introduces a **prototype-based object mining** module, designed to identify **unlabeled** objects across diverse scenes using class-aware prototypes.
- **Second**, a **multi-label cooperative refinement** module is proposed, which integrates pseudo labels with prototype labels to reduce missed detections and improve accuracy.
- **Finally**, the overall **training strategy** is presented.

Methods - Prototype-based object mining



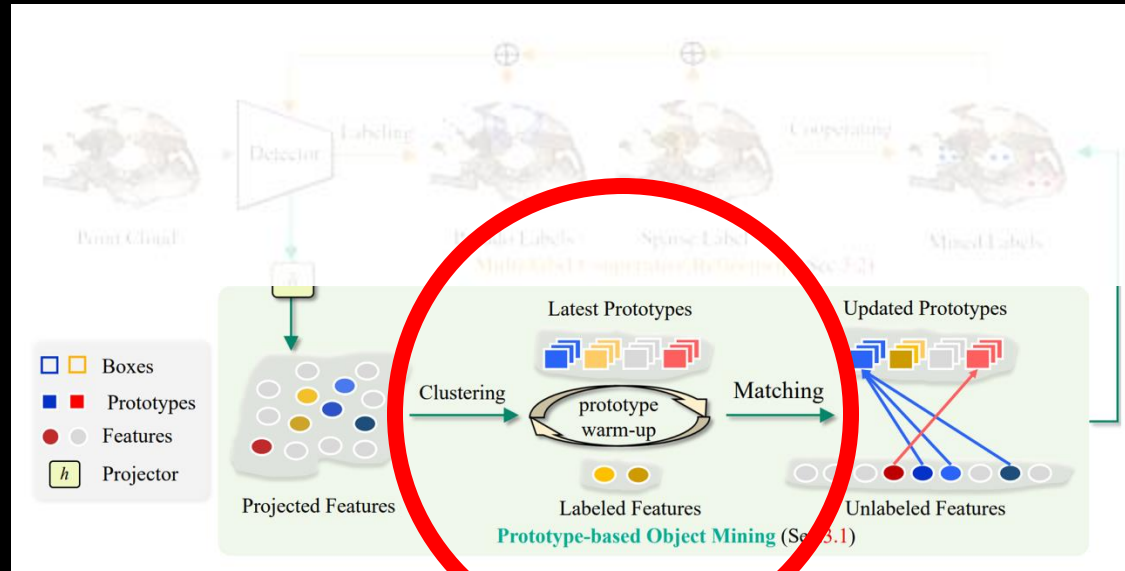
- A prototype-based object mining module is introduced to formulate **unlabeled object discovery** as a **matching** problem between class prototypes and unlabeled features.
→By assigning prototype labels to high-confidence features, this module enables effective mining of unlabeled objects across diverse scenes.

Methods – Projected Features



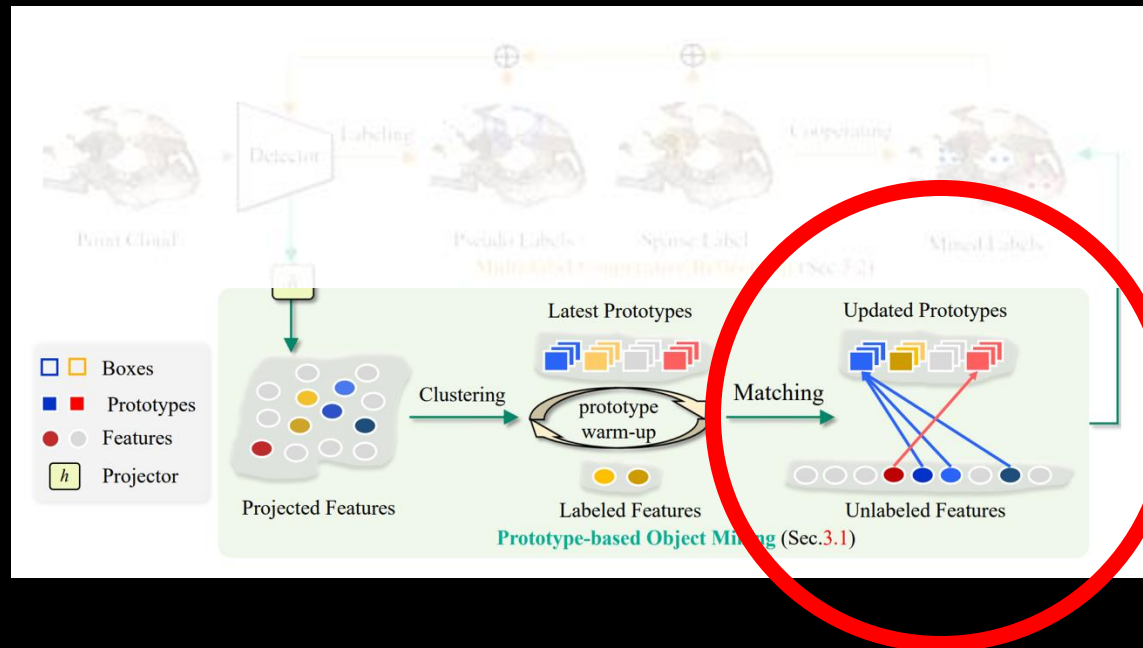
- Proposal features from the detector $X \in \mathbb{R}^{N \times C}$ are passed through a projector composed of several MLP layers to obtain the **projected features** $F \in \mathbb{R}^{N \times C}$, which are used for **prototype learning**.

Methods – Labeled Features



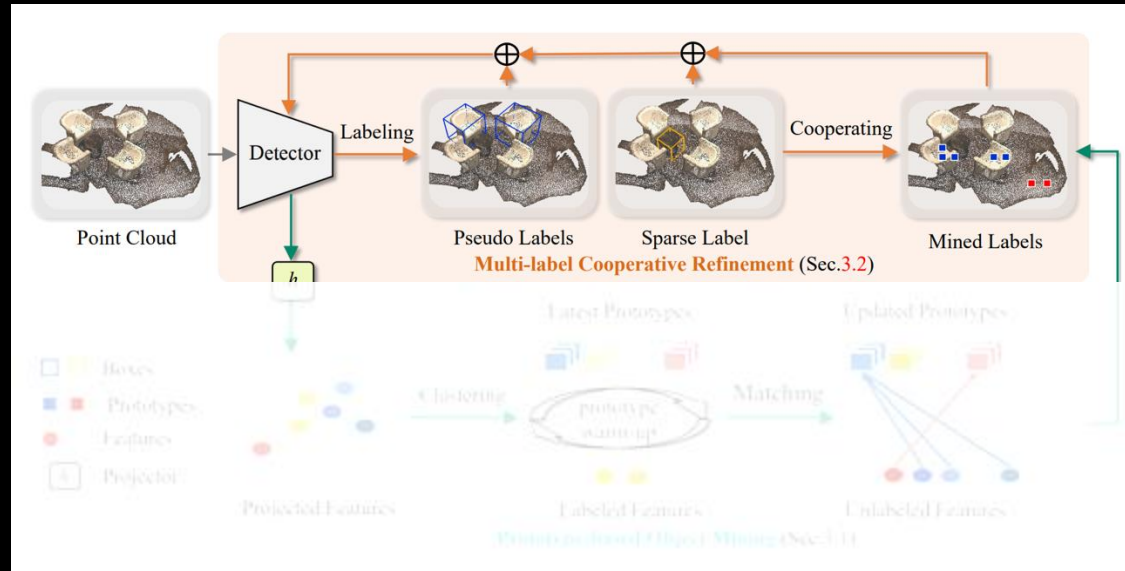
- **Labeled features** F_k are extracted from projected features F using class-aware masks M_k , which identify the locations of sparse ground truth labels for each category.

Methods – Unlabeled Features



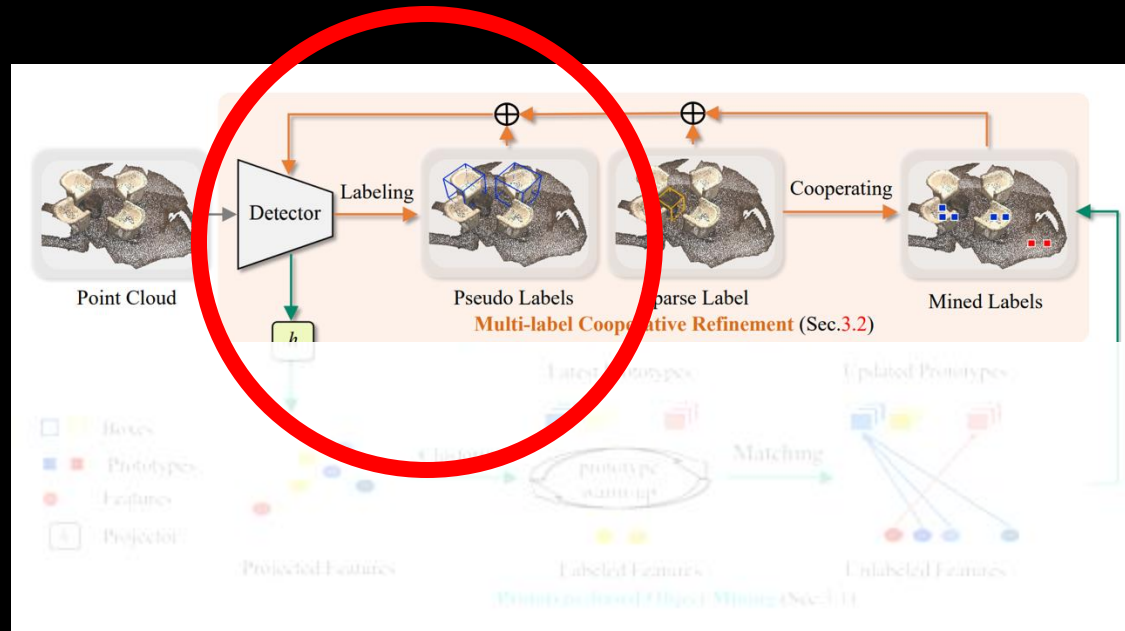
- **Unlabeled features** are the detector's projected features $F \in \mathbb{R}^{N \times C}$ that do not overlap with labeled objects.
- These features are **matched to learned class prototypes**, and category labels are assigned based on the combined similarity and classification confidence.

Methods - Multi-label Cooperative Refinement



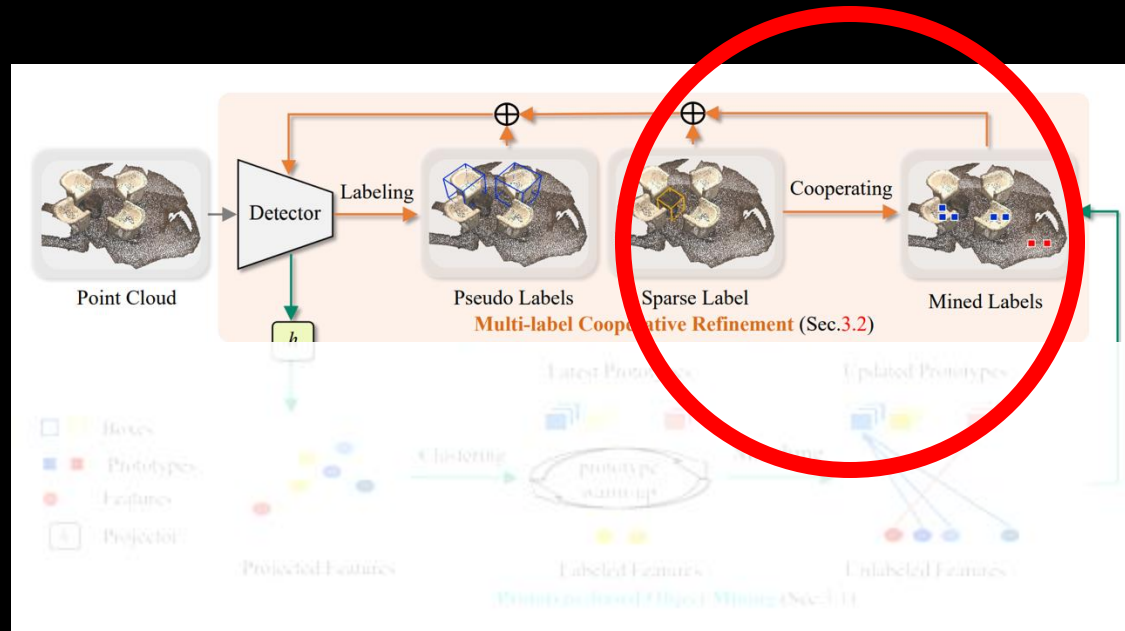
- A **multi-label cooperative refinement** module to effectively recover missed detections through pseudo label **quality control** and prototype label cooperation

Methods – Iterative Pseudo Labeling



- **Score Filter:** remove low-confidence predictions using a threshold α_{cls}
- **IoU Filter:** avoid redundant pseudo boxes using an IoU threshold α_{iou}
- **Collision Filter:** remove pseudo boxes that overlap too much with ground truth using α_{col}

Methods - Prototype Label Cooperating



- Prototype Label Cooperating **recovers missed detections** by assigning class labels to **high-confidence foreground features** that are not covered by sparse or pseudo labels, using class prototype matching.

Methods – Training strategy

Stage 1: an initial detector train

$$\mathcal{L}_{stage1} = \mathcal{L}_{det} + \mathcal{L}_{pcon} + \mathcal{L}_{pcls}$$

- $\mathcal{L}_{det}$: detection loss
- $\mathcal{L}_{pcls}$: prototype classification loss (Focal loss)
- $\mathcal{L}_{pcon}$: contrastive loss (Info-NCE)

Stage 2: pseudo-labels for unlabeled data train

$$\mathcal{L}_{stage2} = \mathcal{L}_{stage1} + \mathcal{L}_{ref}$$

- $\mathcal{L}_{ref}$: detection loss using pseudo labels

Experiments

- Evaluated on **ScanNet V2**, **SUN RGB-D** (indoor) and **KITTI** (outdoor).
- **TR3D** is used for indoor, **Voxel-RCNN + CenterHead** for outdoor.



ScanNet V2



SUN RGB-D



KITTI

Experiments - Indoor

Methods	Present at	Paradigm	ScanNet V2		SUN RGB-D	
			mAP@0.25	mAP@0.5	mAP@0.25	mAP@0.5
FCAF3D [27]	ECCV	Fully Supervised	70.7	56.0	63.8	48.2
TR3D [28]	ICIP		72.0	57.4	66.3	49.6
TR3D [28]	ICIP	Sparse Supervised	37.6	21.8	53.9	36.3
Co-mining [38]	AAAI		43.3	26.4	55.9	39.0
SparseDet [26]	ICCV		46.0	28.2	56.7	38.8
CoIn [42]	ICCV		38.3	23.8	54.8	37.1
CPDet3D (ours)	-		56.1	40.8	60.2	43.3

- **Table 1** shows the comparison with state-of-the-art methods on indoor datasets under the sparse supervision setting.

This work achieves average improvements of **11.3 points on ScanNet V2** and **4.0 points on SUN RGB-D** compared to SparseDet.

Compared to fully supervised methods, it reaches **78% of the fully supervised performance on ScanNet V2** and **90% on SUN RGB-D**

Experiments - Outdoor

Methods	Present at	Paradigm	Car-3D AP (R40)			Car-BEV AP (R40)		
			Easy	Moderate	Hard	Easy	Moderate	Hard
CenterPoint [47]	CVPR	Fully Supervised	89.0	80.5	76.5	92.9	89.0	87.5
Voxel-RCNN [5]	AAAI		92.3	85.2	82.8	95.5	91.2	88.9
Voxel-RCNN [5]	AAAI	Sparse Supervised	72.5	54.9	44.8	83.6	71.4	57.7
SS3D* [20]	CVPR		88.8	78.5	76.9	-	-	-
CoIn [42]	ICCV		84.5	68.4	58.0	92.3	81.0	70.2
CoIn++ [42]	ICCV		92.0	79.5	71.5	96.1	88.8	82.5
CPDet3D (ours)	-		94.1	82.2	72.6	96.2	91.8	83.9

- **Table 2** presents the comparison with state-of-the-art methods on the outdoor dataset under the sparse supervision setting.

This work outperforms **CoIn++** at the moderate difficulty level, achieving a **2.7-point improvement in Car-3D AP** (82.2 vs. 79.5) and a **3.0-point improvement in Car-BEV AP** (91.8 vs. 88.8).

Experiments - Ablation

Number	PLM	CPC	MCR	mAP@0.25	mAP@0.5
1				37.6	21.8
2	✓			38.0	23.5
3	✓	✓		39.3	26.5
4			✓	51.9	32.3
5	✓		✓	54.2	38.3
6	✓	✓	✓	56.1	40.8

- **Table 5.** Ablation study of key components including Prototype-based Label Mining, Class Prototype Clustering, and Multi-label Cooperative Refinement.

PLM alone improves performance by ~3.3/2.1 points, showing effectiveness of prototype-based label mining.

CPC further enhances clustering quality and overall performance.

MCR brings the largest gain in mAP@0.5 (+7.3), showing its importance for recovering missed detections.

Conclusion

- This work proposes a **unified sparse supervised 3D object detection** method applicable to both indoor and outdoor environments.
- A **prototype-based object mining** module was introduced, which performs class-aware prototype clustering from limited labeled data and assigns labels to unlabeled objects via prototype similarity matching.
- A **multi-label cooperative refinement** module integrates high-quality pseudo labels with prototype labels to recover missed detections, while avoiding complex iterative training schemes.

Questions?